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Physiotherapy Theory and Practice
An International Journal of Physical Therapy

ISSN: 0959-3985 (Print) 1532-5040 (Online) Journal homepage: www.tandfonline.com/journals/iptp20

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To cite this article: Pooja Bhati & M. Ejaz Hussain (2023) Impact of resistance training on muscle fatigue in type 2 diabetes mellitus patients during dynamic fatigue protocol, Physiotherapy Theory and Practice, 39:1, 26-38, DOI: 10.1080/09593985.2021.2001882

To link to this article: https://doi.org/10.1080/09593985.2021.2001882

Published online: 08 Nov 2021.

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PHYSIOTHERAPY THEORY AND PRACTICE
2023, VOL. 39, NO. 1, 26–38
https://doi.org/10.1080/09593985.2021.2001882

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RESEARCH REPORT

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Impact of resistance training on muscle fatigue in type 2 diabetes mellitus patients during dynamic fatigue protocol

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ABSTRACT

Background: Type 2 diabetes mellitus (T2DM) is often accompanied with several neuromuscular complications of which muscle fatigue is one of the scantily studied and less-explored complication of this disease.

Objective: To investigate the effects of resistance training (RT) on muscle fatigue markers from electromyography (EMG) and capillary blood in T2DM patients.

Methods: Forty T2DM patients were randomized to either RT (n = 20) or control group (n = 20). They were assessed for EMG indices of muscle fatigue along with blood lactate (at three time points: $Lactate_{pre}$, $Lactate_{peak}$, $Lactate_{post}$) during a dynamic fatigue protocol. Dynamic fatigue protocol consisted of 5 sets of leg press exercise at 10 repetition maximum intensity. RT group performed moderate intensity (65%–75% 1 repetition maximum) RT 3 times/week for 12 weeks. Control group followed usual routine.

Results: Findings revealed that EMG amplitude (root mean square) and Dimitrov's muscle fatigue index (FI_{nsmk}) decreased significantly while median frequency (MF) increased significantly during the dynamic fatigue protocol for both vastus medialis (Root mean square, $p = .02$; FI_{nsmk}, $p = .001$; MF, $p < .001$) and vastus lateralis muscle (Root mean square, $p = .04$, FI_{nsmk}, $p = .01$; MF, $p < .001$) post-RT. Blood lactate responses did not change after RT ($Lactate_{pre}$, $p = .55$; $Lactate_{peak}$, $p = .91$; $Lactate_{post}$, $p = .33$).

Conclusions: Findings of the present study illustrated that moderate intensity RT led to a significant reduction muscle fatigue in knee extensor muscles of T2DM patients. These results reinforce the utilization of RT in patients with T2DM.

ARTICLE HISTORY
Received 15 June 2020
Revised 7 July 2021
Accepted 2 October 2021

KEYWORDS
Exercise; fatigue; muscle; strength; diabetes

Introduction

Patients with type 2 diabetes mellitus (T2DM) commonly experience muscle fatigue, which may be incapacitating and adversely affect self-care regimens. Peripheral muscle fatigue can be defined as "the decline in muscle tension (force) capacity with repeated stimulation" (Kalra and Sahay, 2018. Muscle fatigue in T2DM patients primarily develops due to an increase in the percentage of type II fibers, and a reduction in the percentage of fatigue resistant type 1 fibers in the skeletal muscles accompanied with other metabolic changes (Nyholm et al., 1997; Oberbach et al., 2006). Lifestyle intervention such as exercise is considered to be an integral part of the comprehensive management of T2DM. Particularly, resistance training (RT) has been recently acclaimed to be of great benefit for improving important clinical endpoints in T2DM patients. Although RT at both low (Álvarez, Ramírez-Campillo, Ramírez-Vélez, and Izquierdo, 2017) and high intensity (Dunstan et al., 2002) has shown meaningful improvements in glycemic control but moderate intensity depicted better tolerance rates (Ahmad, 2019). RT has been approved to be the choice of intervention for maintaining optimal glycemic control in T2DM patients; however, its role in managing muscle fatigue is not yet elucidated.

Surface electromyography (EMG) is an excellent method with benefits of real-time monitoring of muscle fatigue during the performance of defined work by measuring myoelectric activity (Cifrek, Medved, Tonković, and Ostojić, 2009). During fatigue, a decline in spectral parameters such as mean and median frequency (MF) and an increase in amplitude parameters such as root mean square (RMS) has been observed (Cifrek, Medved, Tonković, and Ostojić, 2009). Recently, a new highly sensitive spectral index of muscle fatigue (i.e. Dimitrov's muscle fatigue index (FI_{nsmk})) has been proposed which has shown good sensitivity and specificity to detect muscle fatigue during dynamic contractions (Arabadzhiev, Dimitrov, and Dimitrova, 2005;

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Dimitrova, Hogrel, Arabadzhiev, and Dimitrov, 2005). During exercise, when aerobic processes are not able to meet energy demands, there occurs a shift to anaerobic processes, end product of which is lactic acid. When lactate clearance is not equivalent to lactate production, lactic acidosis commences, and its level in the blood is commonly used as a biomarker for evaluating muscle fatigue (Gosker and Schols, 2008). Alterations in surface EMG indices of muscle fatigue and blood lactate responses during exercise have been observed in T2DM patients (Bhati, Singla, Masood, and Hussain, 2021).

At present, though there are no standard recommendations for the management of muscle fatigue, however, there are some supportive treatment strategies which have shown promising results. Research has proposed a few nonspecific treatments in the form of: synthetic products (i.e. amphetamine and caffeine); herbal products (i.e. American ginseng and rhodiola rosea); and nutritional supplements (i.e. vitamins and minerals) (Wan et al., 2017). Interestingly, exercise training has also been recognized as a potential treatment strategy for muscle fatigue (Bogdanis, 2012). Despite the existence of plenty of literature regarding the presence of muscle fatigue in T2DM that too early in the course of the disease (Orlando et al., 2017), there are no uniform management strategies for muscle fatigue in this patient population. RT could be considered a potential treatment for muscle fatigue in T2DM due to its profound positive impact on the neuromuscular system (Orlando et al., 2016). Therefore, the purpose of the present study was to investigate the effects of RT on muscle fatigue markers evaluated from EMG and capillary blood in T2DM patients. Muscle strength and muscle fatigue are closely related to each other and there is concomitant reduction in muscle strength along with deterioration in muscle fatigue in T2DM patients (Orlando et al., 2016); therefore, we also aimed to investigate the effect of moderate intensity RT of 12 weeks on upper and lower body muscle strength in T2DM patients. We hypothesized that moderate intensity RT of 12 weeks will lead to a significant improvement in muscle fatigue and strength parameters in T2DM patients.

Methods

The present study was prospectively registered under Clinical Trial Registry, India on 17–04-2018 (registration number: CTRI/2018/04/013321). The present study is reported in accordance with the CONSORT statement.

Participants

Patients with T2DM (HbA1c > 6.5%) (American Diabetes Association, 2016) with age between 35 and 59 years were recruited into the study. T2DM patients with any severe cardiovascular disease (Criteria Committee of the New York Heart Association, 1994); uncontrolled hypertension [systolic/ diastolic blood pressure (SBP/DBP) > 165/95 mmHg]; or with any clinical orthopedic problem that could limit patient's ability to participate in exercise testing or training without inducing pain, involved in any structured exercise training from last 6 months were excluded.

Study protocol

The present study was a parallel group single blinded (i.e. blinding of outcome assessor) randomized control trial conducted at Center for Physiotherapy and Rehabilitation Sciences (CPRS), Jamia Millia Islamia (JMI) after taking approval from the Institutional Ethics Committee, JMI. Research procedures were carried out in accordance with the Declaration of Helsinki, 1964 (World Medical Association, 1964). After explaining the study procedures to the participants, a written informed consent was obtained from them for their participation into the study. Based on the pre-decided eligibility criteria, a total of 40 T2DM patients were recruited. Participants were provided adequate rest and were completely hydrated prior to the assessments. The testing was conducted over a period of two days. On day 1, after recording the demographic data, 10 repetition maximum (RM) load at which the fatigue protocol has to be performed was determined for leg press exercise using a sequential protocol (Häkkinen et al.,; Willardson and Bressel, 2004). On day 2, after 72 hours of day 1, after maximum isometric voluntary contraction (MVIC) assessment of vastus medialis (VM) and vastus lateralis (VL) muscle, participants were asked to perform a dynamic fatigue protocol at 10 RM load during which EMG for VM and VL muscles and blood lactate levels were recorded. Adequate rest period was provided between MVIC and fatigue testing. All the assessment procedures were conducted between 9.00 and 12.00 am in the morning in order to prevent any diurnal variations in the recordings. Also, the testing conditions were kept similar before and after the intervention in order to prevent the influence of any environmental confounders. Thereafter, 40 T2DM patients were randomly allocated to either RT (RT) (n = 20) or control group (n = 20) using computer generated randomization (Figure 1). Participants were allocated into groups using sequentially numbered, sealed, opaque envelopes

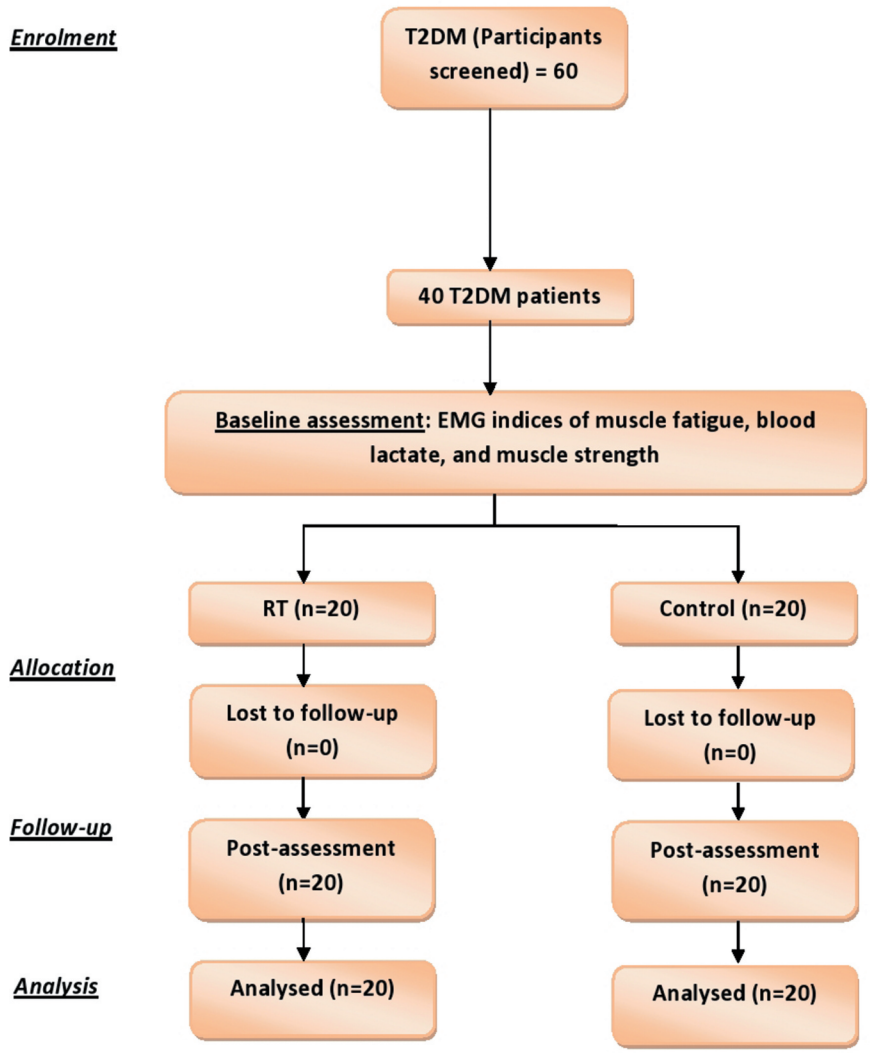

Figure 1. Participant flow through the study. RT: resistance training; T2DM: type 2 diabetes mellitus; EMG: electromyography.

in a 1:1 allocation ratio. Participants in the RT group received moderate intensity whole body RT 3 times/week for 12 weeks while the control group was asked to follow their usual routine. Outcome variables were again assessed after a period of 12 weeks.

Outcome variables

Muscle strength assessment

Repetition Maximum (RM) testing. Muscle strength was measured for both upper [1-RM bench press (BP)] and lower [1-RM leg press (LP)] body musculature, using bench press (Welcare, India) and leg press machines (eight Station Multigym, Fitness World, India). After an adequate warm-up, participant was instructed to perform a set of no more than nine repetitions with a slightly greater weight. The resistance was progressively increased if the participant was able to perform a set of more than nine repetitions until a set of only nine or fewer repetitions was attained. A rest period of 2 min was allowed between each attempt (Hameed et al., 2012). 1RM was estimated using Brzycki 1-RM prediction equation (Brzycki, 1993), based on the resistance and repetitions recorded.

$$1RM = Weight \div [1.0278 - (0.0278 \times number of repetitions)]$$

Maximum Voluntary Isometric Contraction (MVIC) testing. For MVIC assessment, participants were in a seated position on the knee extensor chair with back adjustments and support. The knee and hip joint were placed in 90°of flexion during the MVIC testing. The trunk and thigh were firmly strapped to the chair with a seatbelt. EMG signals were collected for 5 seconds while the participants maintained their knee joints at an angle of 90° against the resistance offered by a trained physiotherapist above the ankle joint. Three subsequent

trials of MVIC were performed for each muscle with a rest period of 2 min between the trials. An average RMS value of the 3 MVIC trials represented MVIC of the concerned muscle. Verbal encouragement was provided to ensure that the participants reached their maximal performance in each trial.

Muscle fatigue assessment

Skin preparation and electrode placement

The skin at the attachment sites was shaved, abraded and cleaned using alcohol swabs as per the guidelines proposed by Surface Electromyography for the noninvasive Assessment of Muscles (SENIAM) (Hermens et al., 1999). Bipolar surface electrodes (AgCl) (Medico Electrodes International LTD, India) were utilized for recording surface EMG from VM and VL muscles of the right leg. Electrodes were arranged in parallel to the orientation of the muscle fibers with an interelectrode distance of 20 mm. For VM, electrodes were placed at "the eight-tenths of the distance along the line connecting anterior superior iliac spine (ASIS) and the superior medial margin of the patellar bone." For VL, electrodes were placed at "the two-thirds of the distance along the line connecting ASIS and the lateral side of the patellar bone" (Hermens et al., 2000). The reference electrode was placed over the patella (Balshaw et al., 2017).

Dynamic fatigue protocol

Ten RM (intensity at which the fatigue protocol was executed) testing was performed 72 h before the actual dynamic fatigue protocol testing. Participants were asked to sit on the bilateral leg press machine and perform leg extensions against the load determined by a trained physiotherapist. Resistance was increased at each stage until a load at which the participant was unable to perform the eleventh repetition and this load was considered as the 10 RM load (Willardson and Bressel, 2004). The dynamic fatigue protocol constituted 5 sets of 10 repetitions each (2 min rest between the sets) at the 10 RM load. Patients were instructed to follow the rhythm of a metronome (40 bpm) during the execution of repetitions (Izquierdo et al., 2011). EMG was simultaneously recorded during the fatigue protocol from VM and VL muscles. Surface EMG has demonstrated good test–retest reliability during dynamic contractions (Larsson, Karlsson, Eriksson, and Gerdle, 2003).

Signal acquisition and analysis

All the data was recorded on Lab Chart software version 8, AD instruments, Australia. EMG signals were obtained at a sampling rate of 1000 Hz and processed by a 8–500 Hz band-pass filter and an amplifier with a common mode rejection ratio > 100 dB. All 10 EMG segments for each set were extracted for calculating EMG fatigue indices. Fast Fourier Transform (FFT) was performed which decomposed the EMG signal into frequencies for calculating MF. EMG amplitude was obtained in the form of RMS value for each contraction (Lab Chart 8, AD instruments). Another spectral parameter of muscle fatigue (i.e. Dimitrov's muscle fatigue index (FI_{nsmk})) which consists of a ratio between different spectral moments calculated over the power spectral density obtained via FFT, was calculated using the Matlab software (The MathWorks, Inc, Natick, MA) (Izquierdo et al., 2011). The FI_{nsmk} was calculated as the ratio between the spectral moments of order (−1) and order k:

$$FI_{nsmk} = \frac{{}_{f1}f1f-1PS(f).}{{}_{f1}f1fkPS(f).df}$$

Where, order k was set as 2, 3, 4, or 5, PS(f) was the spectral power for the current frequency such that $f1$ and $f2$ define the spectral bandwidth with $f1$ = 8 Hz and $f2$ = 500 Hz being the high and low-pass frequencies of the amplifier respectively. Average RMS, MF and FI_{nsmk} were obtained for each set (set 1 to set 5) as measures of muscle fatigue.

Blood lactate assessment

After cleaning the skin with ethanol, capillary samples were obtained from the participant's finger for the assessment of blood lactate using a blood lactate analyzer (Lactate Scout+, EKF Diagnostics, UK) during the fatigue protocol. Blood samples were obtained before the 1st set and immediately after the 3rd and 5th sets in order to obtain $Lactate_{pre}$, $Lactate_{mid}$, and $Lactate_{post}$ values (Izquierdo et al., 2011).

Resistance training

Exercise sessions were conducted at the Human Performance Laboratory, CPRS, JMI under close supervision of a trained physiotherapist. Moderate intensity progressive RT (65%-75% of 1 RM) was performed 3 times/week for 12 weeks. Two to three sets of 8–12 repetitions (2–3 min of rest between the sets) were performed for each exercise. Each exercise session lasted for around 60 min. Exercise intensity was progressed gradually from 65% of 1 RM to 75% of 1 RM over a period of 12 weeks. For the initial 4 weeks, intensity was kept at 65% of 1 RM. It was progressed to 70% of mid-study 1RM at the end of 4th week and then to 75% of 1RM calculated at 8th

week. Warm up and cool down consisted of 10 min of cycling on self-selected pace and low load prolonged stretches (two to three repetitions) of the exercising muscles. Resistance exercises were executed with the help of weight machines and free weights in a consistent order: Leg press, abdominal curl, lateral pulldown, seated row, bicep curl, butterfly, bench press, knee extension, overhead triceps extension, and standing leg curl (Colberg et al., 2010). Control group was asked to follow conventional medical treatment and daily routine activities and was instructed to not engage in any structured exercise training program.

Statistical analysis

Data analysis was performed on SPSS software version 21 (SPSS Inc., Chicago, IL, USA). Demographic and clinical characteristics were compared between the RT and control group using independent *t* test. Chi-square test was used to compare gender differences between the groups. A 2 × 2 repeated measures analysis of variance (ANOVA) was used to analyze blood lactate, glycemic control (HbA1c) and muscle strength between the two groups (RT versus control) at two time points (baseline and 12th week). 2 × 2 × 5 repeated measures ANOVA was employed to examine differences in EMG indices between the two groups (RT versus control) at two time points (baseline and 12th week) across the five sets. Analysis of covariance (ANCOVA) was applied for variables which showed significant differences at baseline between the two groups. Significance level was set at $p < .05$ for the present study.

Results

A post-hoc power analysis performed on all three EMG indices of muscle fatigue (RMS, MF and FI_{nsmk}) for both the muscles (VM and VL) revealed that the present study had adequate power (ranging from 0.83 to 0.99) with effect sizes ranging from 0.36 to 0.88 and a sample size of 40 T2DM patients was sufficient to test the study hypothesis.

All 40 T2DM patients recruited in this study completed the study procedures and there were no drop-outs in this study (Figure 1). Demographic and outcome variables were statistically similar between the RT and control group except for 1 RM BP (Table 1), which was analyzed using ANCOVA. Upper body (12.2 ± 8.72 to 16.6 ± 9.82 kg) and lower body strength (43.3 ± 14.90 to 51.4 ± 15.35 kg) improved significantly post-RT. A significant time ($p = .003$) and group×time interaction ($p < .001$) was observed for 1RM LP and a significant group effect ($p = .005$) was observed for 1RM BP.

A significant time ($p < .001$) and interaction effects ($p = .002$) were also observed for HbA1c after RT, though group effect ($p = .23$) was insignificant for the same. Statistically significant and near significant increase in MVIC of VM [RT, 0.13 ± 0.01 to 0.15 ± 0.01 mV versus Control, 0.12 ± 0.01 to 0.11 ± 0.009 mV, Group: 0.93, Time: 0.05, Interaction: 0.006] and VL muscles [RT, 0.40 ± 0.006 to 0.47 ± 0.08 mV versus Control, 0.34 ± 0.08 to 0.25 ± 0.06 mV; Group: 0.06, Time: 0.78, Interaction: 0.057] was observed, respectively.

Table 2 and Figure 2 shows that the increase in EMG amplitude (RMS_{VM}), declined after RT indicating lesser fatigue development in response to training in VM muscle. Analysis revealed a significant time ($p < .001$), group×time ($p = .02$), sets ($p < .001$), time×sets ($p = .01$), and time×sets×group ($p = .007$) effect for RMS in VM muscle. Similarly, for VL muscle, apart from a significant time effect ($p < .001$), a significant time×sets×group interaction ($p = .04$) was also observed which indicates lesser fatigue development post-RT (Table 3 and Figure 3). A significant time ($p < .001$), group×time ($p < .001$), sets ($p < .001$), time×sets ($p < .001$), and group effect ($p = .002$) were observed for MF in the VM muscle (Table 2 and Figure 2). Similarly, the decline in MFwas found to be reduced after RT for the VL muscle as well and a significant group×time interaction effect ($p < .001$) was observed (Table 3 and Figure 3). Significant decline was observed in FI_{nsmk} in the RT group in contrary to control group which showed an increase in FI_{nsmk} for both VM and VL muscles (Table 2–3 and Figure 2–3). A significant group×time ($p = .001$) and time×sets×group effect ($p = .02$) were observed for FI_{nsmk} of VM muscle (Table 2 and Figure 2) indicating significant effect of RT over control on FI_{nsmk} of the VM muscle. Similarly, a significant group effect ($p = .01$) was observed for FI_{nsmk} indicating a significant decline in FI_{nsmk} of VL muscle after RT over control group (Table 3 and Figure 3).

There was no significant change in blood lactate levels post-RT. Blood lactate responses at baseline and after 12th week in the RT versus control group were: $lactate_{pre}$ (RT: 2.8 ± 1.12–2.7 ± 0.98, Control: 2.8 ± 1.38–2.9 ± 1.53; Group, $p = .77$, Time, $p = .63$; Group×Time, $p = .55$), $lactate_{peak}$ (RT: 3.28 ± 1.50–3.09 ± 1.29, Control: 3.2 ± 1.45 to 3.0 ± 1.44; Group, $p = .84$, Time, $p = .21$; Group×Time, $p = .91$) and

Table 1. Comparison of demographic and outcome variables between the groups at baseline.

Variables	RT (n = 20)	Control (n = 20)	*p* value
	Mean ± SD	Mean ± SD	
Demographic variables			
Age (years)	54.5 ± 6.84	51.0 ± 8.86	0.17
Males (n)	12	12	1.00
Females (n)	8	8	1.00
Weight (kg)	73.0 ± 7.39	73.5 ± 9.26	0.86
Height (m)	161.9 ± 8.71	162.6 ± 7.27	0.76
BMI (kg/m^2)	28.0 ± 3.51	27.8 ± 3.58	0.87
HbA1c	8.4 ± 1.49	8.7 ± 1.98	0.67
Muscle strength			
1RM BP (kg)	12.2 ± 8.72	8.0 ± 4.82	0.01*
IRM LP (kg)	43.4 ± 14.90	41.5 ± 13.97	0.58
EMG indices of muscle fatigue			
RMS_{VM} (mV)			
Set 1	0.13 ± 0.09	0.14 ± 0.07	0.61
Set 2	0.22 ± 0.14	0.17 ± 0.08	0.20
Set 3	0.22 ± 0.11	0.19 ± 0.08	0.41
Set 4	0.23 ± 0.12	0.19 ± 0.09	0.28
Set 5	0.24 ± 0.13	0.20 ± 0.10	0.26
RMS_{VL} (mV)			
Set 1	0.26 ± 0.19	0.22 ± 0.13	0.43
Set 2	0.24 ± 0.17	0.23 ± 0.15	0.81
Set 3	0.30 ± 0.15	0.24 ± 0.13	0.24
Set 4	0.30 ± 0.14	0.26 ± 0.14	0.50
Set 5	0.31 ± 0.05	0.27 ± 0.08	0.89
MF_{VM} (Hz)			
Set 1	54.35 ± 7.54	54.37 ± 6.23	0.99
Set 2	52.92 ± 7.58	53.14 ± 5.41	0.79
Set 3	48.43 ± 7.97	49.01 ± 5.60	0.79
Set 4	43.52 ± 8.30	42.10 ± 4.26	0.85
Set 5	42.58 ± 5.57	43.19 ± 4.06	0.61
MF_{VL} (Hz)			
Set 1	53.03 ± 9.50	54.32 ± 8.18	0.62
Set 2	52.09 ± 0.9	53.14 ± 2.11	0.18
Set 3	53.14 ± 3.16	53.04 ± 1.74	0.90
Set 4	50.51 ± 4.61	51.35 ± 4.45	0.56
Set 5	49.84 ± 3.83	49.88 ± 3.53	0.69
$FInsmk_{VM}$ (Hz^{-k})			
Set 1	4.06 ± 0.92 e-15	5.54 ± 2.14 e-15	0.62
Set 2	4.53 ± 1.16 e-15	3.17 ± 1.02 e-15	0.26
Set 3	1.02 ± 0.20 e-14	1.22 ± 0.46 e-14	0.40
Set 4	2.06 ± 0.50 e-14	1.49 ± 0.34 e-14	0.35
Set 5	2.99 ± 0.82 e-14	1.65 ± 0.37 e-14	0.21
$FInsmk_{VL}$ (Hz^{-k})			
Set 1	3.58 ± 0.67e-15	5.61 ± 1.56 e-15	0.24
Set 2	5.74 ± 1.32 e-15	5.41 ± 1.30 e-15	0.85
Set 3	7.57 ± 1.87 e-15	9.35 ± 2.12 e-15	0.53
Set 4	2.69 ± 0.81 e-14	2.20 ± 0.80 e-14	0.67
Set 5	2.17 ± 0.94 e-14	5.71 ± 1.73 e-14	0.08
Blood lactate			
$Lactate_{pre}$ (mmol/L)	2.84 ± 1.12	2.80 ± 1.38	0.99
$Lactate_{peak}$ (mmol/L)	3.28 ± 1.50	3.21 ± 1.45	0.90
$Lactate_{post}$ (mmol/L)	3.63 ± 1.60	3.78 ± 1.83	0.85

RT: resistance training; M: male: F: female: BMI: body mass index; VM: vastus medialis; VL: vastus lateralis; MVIC: maximum voluntary isometric contraction; RM: repetition maximum; BP: bench press; LP: leg press; mV: millivolts; RMS: root mean square; FInsmk: Dimitrov's muscle fatigue index; MF: median frequency; %: percentage

$lactate_{post}$ (RT: 3.63 ± 1.60–4.07 ± 2.31, Control: 3.7 ± 1.83–5.2 ± 4.17; Group, p = .43, Time, p = .13; Group×Time, p = .33). Percentage changes in EMG parameters and blood lactate after RT are reported in Table 4.

Discussion

Findings suggested that RT led to a significant decline in EMG amplitude, i.e. RMS and FI_{nsmk} and a significant increase in MF which indicates alleviation of muscle fatigue in response to RT. In the present study, a significant increase in muscle strength was also obtained after RT whereas blood lactate responses remained unchanged.

Neuromuscular impairments in DM results from hyperglycemia-induced oxidative stress in diabetic neurons which directly impacts the intrinsic properties of the muscles related to force production (Izquierdo et al., 2009). Several studies have demonstrated that T2DM patients exhibit greater muscle fatigue than their healthy counterparts (Allen, Kimpinski, Doherty, and Rice, 2015; Orlando et al., 2017). We hypothesized that RT of 12 weeks will result in a significant reduction in muscle fatigue in T2DM patients and in accordance with our experimental hypothesis, findings revealed that 12 weeks of RT led to a significant decline in muscle fatigue in knee extensor muscles of T2DM patients. There was a significant decline in RMS, and FI_{nsmk} and a significant increase in MF values across the sets post-RT which indicates lesser fatigue development after training (Tables 2–3 and Figures 2–3). Izquierdo et al. (2009); Izquierdo et al. (2011) investigated the effects of 7 weeks of progressive RT on EMG indices of muscle fatigue during a dynamic fatigue protocol in healthy men. Findings of both the studies (Izquierdo et al., 2011, 2009) revealed that RT did not lead to any change in surface EMG spectral (MF and FI_{nsmk}) or amplitude parameters (MAV). Differences in the findings of these studies (Izquierdo et al., 2011, 2009) and the present study could be attributed to the differences in the duration of intervention (7 versus 12 weeks) and the population (healthy versus T2DM) studied. It may be speculated that T2DM patients might be more responsive to improvements in response to RT than healthy individuals owing to multiple neuromuscular deficiencies in this patient population (Orlando et al., 2016). Moreover, training period of 7 weeks may be considered insufficient to elicit changes in surface EMG indices of fatigue and indicates that neuromuscular adaptations are only detectable by surface EMG after a sufficient duration of training (i.e. 12 weeks). In accordance with the findings of present study, Oliveira and Goncalves (2009) found a significant decrease in linear regression slope of

Table 2. Electromyographic indices of muscle fatigue for vastus medialis muscle (across five sets) before and after resistance training in T2DM patients.

Variables		RT (n = 20)	Control (n = 20)	*p*-values						
		Mean ± SD/SE	Mean ± SD/SE	Time	Time×Group	Sets	Sets×Group	Time×Sets	Time×Sets×Group	Group
RMS (mV)				**<0.001***	**0.02***	**<0.001***	0.79	**0.01***	**0.007***	0.92
Set 1	Pre	0.13 ± 0.09	0.14 ± 0.07							
	Post	0.11 ± 0.07	0.12 ± 0.04							
Set 2	Pre	0.22 ± 0.14	0.17 ± 0.08							
	Post	0.12 ± 0.06	0.15 ± 0.07							
Set 3	Pre	0.22 ± 0.11	0.19 ± 0.08							
	Post	0.13 ± 0.07	0.16 ± 0.07							
Set 4	Pre	0.23 ± 0.12	0.19 ± 0.09							
	Post	0.13 ± 0.07	0.16 ± 0.07							
Set 5	Pre	0.24 ± 0.13	0.20 ± 0.10							
	Post	0.14 ± 0.09	0.16 ± 0.08							
MF (Hz)				**<0.001***	**<0.001***	**<0.001***	0.99	**<0.001***	0.71	**0.002***
Set 1	Pre	54.35 ± 7.54	54.37 ± 6.23							
	Post	56.71 ± 6.24	52.84 ± 7.21							
Set 2	Pre	52.92 ± 7.58	53.14 ± 5.41							
	Post	55.25 ± 4.19	50.33 ± 5.16							
Set 3	Pre	48.43 ± 7.97	49.01 ± 5.60							
	Post	54.24 ± 3.54	49.98 ± 3.96							
Set 4	Pre	43.52 ± 8.30	42.10 ± 4.26							
	Post	52.44 ± 4.41	48.94 ± 7.09							
Set 5	Pre	42.58 ± 5.57	43.19 ± 4.06							
	Post	53.24 ± 3.25	47.78 ± 3.71							
FI_{nsmk} (Hz^{-k})				0.22	**0.001***	**<0.001***	0.73	0.53	**0.02***	0.26
Set 1	Pre	4.06 ± 0.92 e-15	5.54 ± 2.14 e-15							
	Post	3.84 ± 1.16e-15	5.98 ± 2.10e-15							
Set 2	Pre	4.53 ± 1.16 e-15	3.17 ± 1.02 e-15							
	Post	1.88 ± 0.17e-15	5.50 ± 1.49e-15							
Set 3	Pre	1.02 ± 0.20 e-14	1.22 ± 0.46 e-14							
	Post	3.84 ± 0.76e-15	1.46 ± 0.38e-14							
Set 4	Pre	2.06 ± 0.50 e-14	1.49 ± 0.34 e-14							
	Post	1.01 ± 0.24 e-14	1.77 ± 0.34e-14							
Set 5	Pre	2.99 ± 0.82 e-14	1.65 ± 0.37 e-14							
	Post	9.38 ± 0.25 e-15	3.29 ± 1.06 e-14							

RT: resistance training; SD: standard deviation; RMS: root mean square; MF: median frequency; FI_{nsmk}: Dimitrov's muscle fatigue index; mV: millivolt; data are presented as mean ± standard deviation for root mean square and median frequency and as mean ± standard error for FInsmk; *significant difference

RMS and a significant increase in the slope and intercept of MF after RT in healthy males suggesting lesser fatigue development following RT. Two more studies indicated the same after RT of 8 weeks in young males with a significant increase in muscle endurance during a submaximal task performed till momentary failure (Hong, Hong, and Shin, 2014; Schoenfeld et al., 2019). Increase in fatigue tolerance observed in the present study could be attributed to local muscular factors, such as increased capillarization and myofiber hypertrophy (Andersen et al., 2014) and a shift toward an increased proportion of fatigue-resistant fast-twitch fibers resulting in a more fatigue-resistant muscle fiber profile post-RT (Farup et al., 2014). In addition, RT has also shown to enhance glucose metabolism and increase oxidative enzymes (Groennebaek and Vissing, 2017) and reduce glycolytic muscle enzymes (Tesch, Komi, and Häkkinen, 1987), which might have reduced fatigue in the present experiment. Therefore, in summary, changes in the muscle fiber type and distribution, enhanced glucose metabolism, and an increase in mitochondrial oxidative enzymes may explain reduced muscle fatigue post-RT in the present study.

Findings of the present study revealed that there were no significant differences in the blood lactate response between the groups post-RT. In contrary to the findings of present experiment, previous studies (Izquierdo et al., 2011, 2009) have found a significant rise in blood lactate post-RT which indicates development of greater metabolic stress after RT. The contrasting results of the present and previous investigations indicate that healthy and diseased populations respond differently to RT and that diseased population (i.e. T2DM) may utilize more aerobic glycolysis than anaerobic glycolysis post-RT and this metabolism is probably reversed in healthy individuals. In accordance with our findings, a few studies (Hurley et al., 1984; Kim et al., 2011) have illustrated that exercise of short (6 weeks) and long duration (12 weeks) reduces blood lactate levels and the expression of lactate metabolic proteins (i.e. monocarboxylate transporter (MCT) proteins –

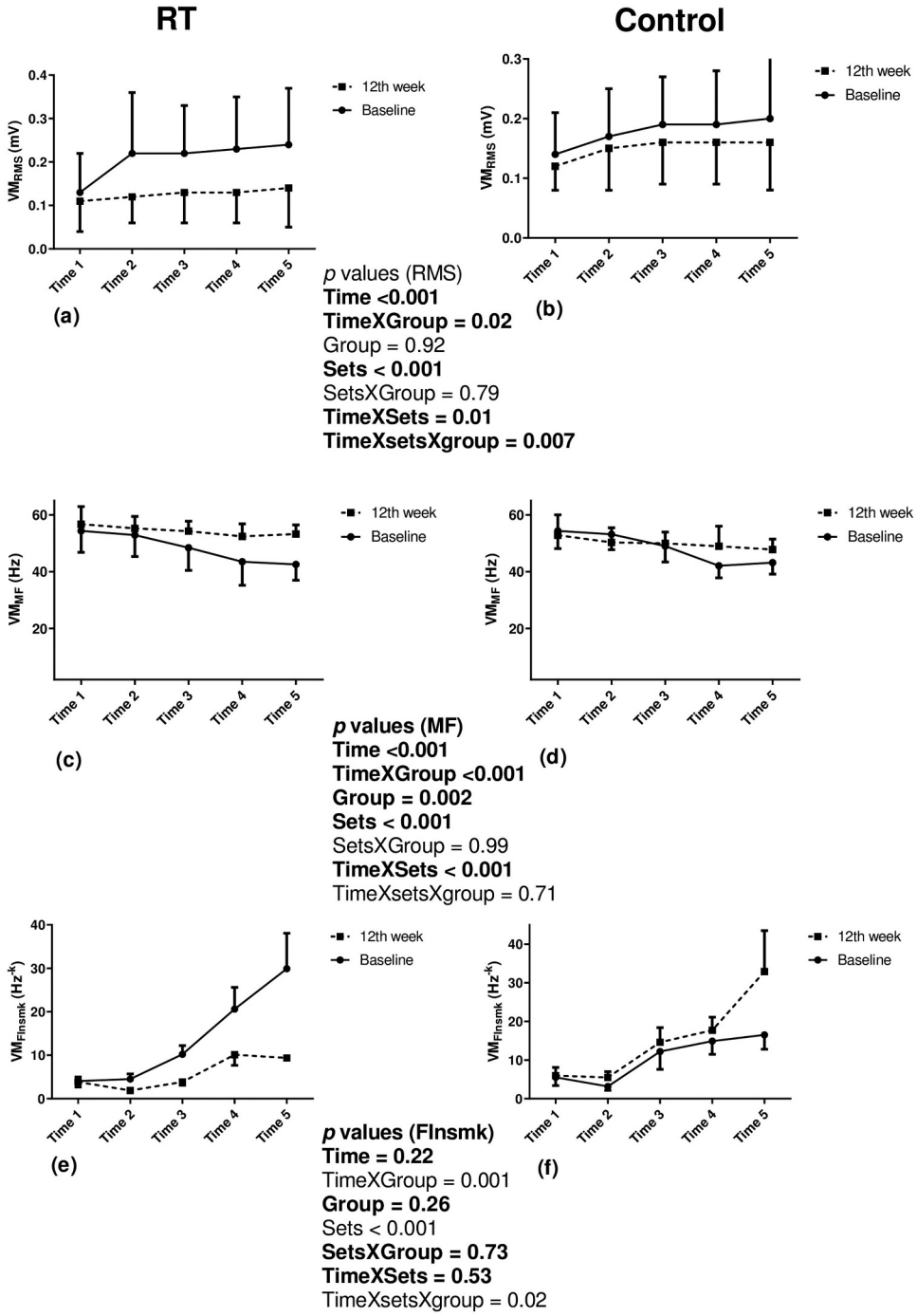

Figure 2. Electromyographic indices of muscle fatigue for vastus medialis muscle before and after 12 weeks of resistance training. RT: resistance training; VM: vastus medialis; RMS: root mean square; MF: median frequency; FI_{nsmk}: Dimitrov's muscle fatigue index; decline in RMS and FI_{nsmk} and an increase in MF could be seen in the RT group indicating reduction in muscle fatigue post-RT.

MCT-1 and MCT-4). MCT-1 is a monocarboxylate transporter protein. Increased MCT-1 expression indicates enhanced lactate transport across the membrane in human skeletal muscle (Juel and Halestrap, 1999). Nonetheless, the only study on T2DM patients revealed that RT of 6 weeks increases the content of MCT-1 in the skeletal muscles of these patients (Juel, Holten, and Dela, 2004). No change in blood lactate responses post-RT in the present study indicates that metabolic demands/lactate metabolism of the fatigue protocol remains the same at baseline and post-training since improvement in EMG indices was not accompanied by changes in blood lactate levels. Nevertheless, it must be noted that lactate concentration in blood gives an estimation of global fatigue of the organism

Table 3. Electromyographic indices of muscle fatigue for vastus lateralis muscle (across five sets) before and after resistance training in T2DM patients.

Variables RMS (mV)		RT (n = 20) Mean ± SD/SE	Control (n = 20) Mean ± SD/SE	*p*-values						
				Time	Time×Group	Sets	Sets×Group	Time×Sets	Time×Sets×Group	Group
Set 1	Pre	0.26 ± 0.19	0.22 ± 0.13	**<0.001***	0.69	0.61	0.27	0.24	0.04*	0.40
	Post	0.17 ± 0.04	0.25 ± 0.19							
Set 2	Pre	0.24 ± 0.17	0.23 ± 0.15							
	Post	0.18 ± 0.04	0.29 ± 0.25							
Set 3	Pre	0.30 ± 0.15	0.24 ± 0.13							
	Post	0.25 ± 0.19	0.31 ± 0.26							
Set 4	Pre	0.30 ± 0.14	0.26 ± 0.14							
	Post	0.25 ± 0.14	0.33 ± 0.21							
Set 5	Pre	0.31 ± 0.05	0.27 ± 0.08							
	Post	0.28 ± 0.20	0.34 ± 0.29							
MF (Hz)				**0.001***	**<0.001***	**<0.001***	0.61	0.12	0.09	0.16
Set 1	Pre	53.03 ± 9.50	54.32 ± 8.18							
	Post	54.21 ± 3.19	52.79 ± 3.39							
Set 2	Pre	52.09 ± 0.9	53.14 ± 2.11							
	Post	54.04 ± 5.19	48.04 ± 4.13							
Set 3	Pre	53.14 ± 3.16	53.04 ± 1.74							
	Post	51.61 ± 3.11	48.04 ± 4.13							
Set 4	Pre	50.51 ± 4.61	51.35 ± 4.45							
	Post	50.13 ± 2.24	47.63 ± 2.92							
Set 5	Pre	49.84 ± 3.83	49.88 ± 3.53							
	Post	49.87 ± 4.62	45.77 ± 8.97							
FI_{nsmk} (Hz^{-k})				0.66	0.08	**<0.001***	**0.002***	0.57	0.22	**0.01***
Set 1	Pre	3.58 ± 0.67e-15	5.61 ± 1.56 e-15							
	Post	3.97 ± 1.35e-15	9.32 ± 2.18e-15							
Set 2	Pre	5.74 ± 1.32 e-15	5.41 ± 1.30 e-15							
	Post	3.99 ± 1.65e-15	9.91 ± 4.55e-15							
Set 3	Pre	7.57 ± 1.87 e-15	9.35 ± 2.12 e-15							
	Post	5.36 ± 1.97e-15	8.08 ± 1.88e-15							
Set 4	Pre	2.69 ± 0.81 e-14	2.20 ± 0.80 e-14							
	Post	1.29 ± 0.83e-14	1.92 ± 0.81e-14							
Set 5	Pre	2.17 ± 0.94 e-14	5.71 ± 1.73 e-14							
	Post	8.98 ± 3.97e-15	1.16 ± 1.81e-13							

RT: resistance training; SD: standard deviation; VL: vastus lateralis; RMS: root mean square; MF: median frequency; FInsmk: Dimitrov muscle fatigue index; Data are presented as mean ± standard for root mean square and median frequency and as mean ± standard error for FInsmk; *significant difference

(Westerblad, Allen, and Lannergren, 2002) whereas EMG was recorded only for the VM and VL muscles. Thus, fatigue/lactate metabolism might have improved in these muscles, however, despite adequate stabilization; contraction of other muscles apart from VM and VL might have contributed to the blood lactate concentration, thus, not giving a clear picture of RT-induced improvement in lactate metabolism. Moreover, since blood is a big reservoir of metabolites which continuously circulates within the body, lactate accumulated in the blood during continuous exercise gets utilized by organs such as liver, heart, and inactive, and moderately active skeletal muscle for lactate oxidation (Goodwin, Harris, Hernández, and Gladden, 2007). Therefore, lactate concentration in the blood could not completely represent the fatigue accumulated in the muscle.

In addition to muscle fatigue, a significant increase has been observed in upper and lower body strength (1RM BP and 1RM LP) post-RT in the present study. In accordance with the findings of the present investigation, previous research has proved the efficacy of RT for enhancing muscle strength (Andrade et al., 2018). It has been proposed that RT increases muscle strength via induction of muscle hypertrophy and neuromuscular remodeling through phosphatidylinositol 3 kinase (P13k)-Akt-mammalian target of rapamycin (mTOR) pathway (Goodman, 2013). Furthermore, morphological changes in the muscles such as an increase in cross-sectional area, volume, improved configuration of satellite cells along with functional neuromuscular adaptations such as an increase in number and size of motor units, enhanced recruitment, and synchronized firing of motor units are the possible mechanisms responsible for enhancing force producing capacity of the muscles (Goodman, 2013). Significant improvement in maximal muscle strength (1RM and MVIC) along with alleviation in muscle fatigue parameters indicates potential of RT in partly improving functional performance of T2DM patients.

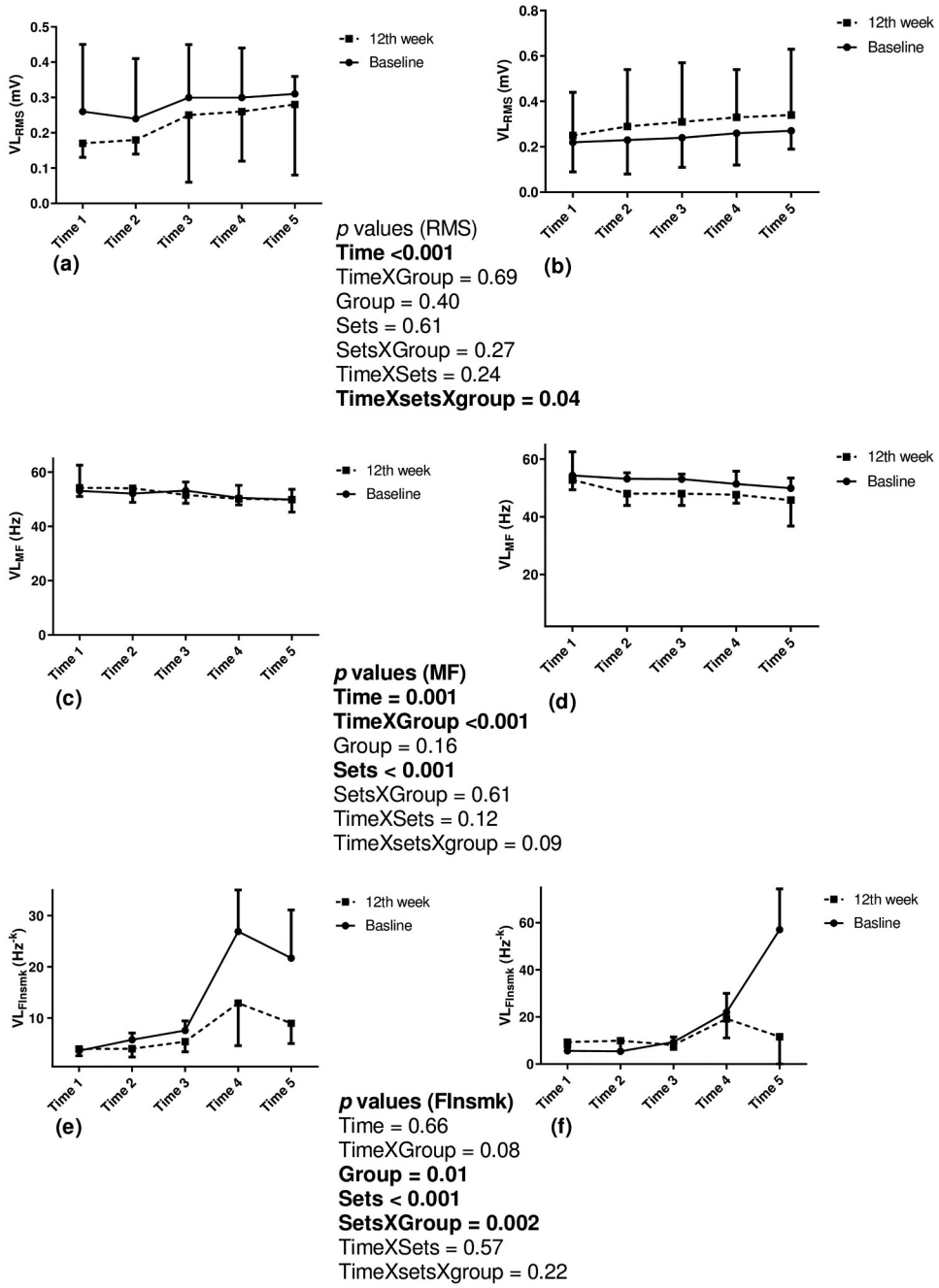

Figure 3. Electromyographic indices of muscle fatigue for vastus lateralis muscle before and after 12 weeks of resistance training. RT: resistance training; VL: vastus lateralis; RMS: root mean square; MF: median frequency; FI_{nsmk}: Dimitrov's muscle fatigue index; decline in RMS and FI_{nsmk} and an increase in MF could be seen in the RT group indicating reduction in muscle fatigue post-RT.

The present study encountered a few limitations. Lactate during the fatigue protocol was assessed in the blood which might have been confounded by the contractions of other adjacent muscles apart from the muscles assessed. Therefore, future research should consider to examine lactate in the concerned muscle in order to clearly understand the mechanism of muscle fatigue reduction by RT. Though every possible measure was taken to securely adhere electrodes to the skin, still, movement of the limbs might have confounded the EMG signal during dynamic fatigue protocol. Functional performance parameters were not assessed

Table 4. Percentage change in EMG parameters and blood lactate after resistance training.

Variables	RT (n = 20) Change (%)	Control (n = 20) Change (%)
EMG parameters		
Vastus medialis		
RMS		
Set 1	−15.3	−14.2
Set 2	−45.4	−11.7
Set 3	−40.9	−15.7
Set 4	−43.4	−15.7
Set 5	−41.6	−20
MF		
Set 1	4.3	−2.8
Set 2	4.4	−5.2
Set 3	11.9	1.97
Set 4	20.4	16.2
Set 5	25.0	10.6
FI_{nsmk}		
Set 1	−5.4	7.9
Set 2	−58.4	73.5
Set 3	−62.3	19.6
Set 4	−50.9	18.7
Set 5	−68.6	99.3
Vastus lateralis		
RMS		
Set 1	−34.6	13.6
Set 2	−25.0	26.0
Set 3	−16.6	29.1
Set 4	−16.6	26.9
Set 5	−9.6	25.9
MF		
Set 1	2.2	−2.8
Set 2	3.7	−9.5
Set 3	−2.8	−9.4
Set 4	−0.7	−7.2
Set 5	0.06	−8.2
FI_{nsmk}		
Set 1	10.8	66.1
Set 2	−30.4	83.1
Set 3	−29.1	−13.5
Set 4	−52.0	−12.7
Set 5	58.6	103.1
Blood lactate		
$Lactate_{pre}$	−4.9	6.4
$Lactate_{peak}$	−5.7	−6.5
$Lactate_{post}$	12.1	39.6

RT: resistance training; RMS: root mean square; MF: median frequency; FInsmk: Dimitrov's muscle fatigue index; %: percentage

in the present study which could have reflected the changes in functional aspect of these patients was another limitation of this study and this should be taken into account by future research.

Conclusions

Findings of the present study suggest that progressive RT of moderate intensity significantly improves muscle strength and muscle fatigue in knee extensor muscles of T2DM patients along with significant improvement in glycemic control. However, it does not significantly changes blood lactate responses during the fatigue protocol. Findings of the present investigation recommends the utilization of RT for alleviating muscle fatigue in patients with T2DM.

Acknowledgments

Authors would like to acknowledge Jamia Millia Islamia for providing logistic help during the study. A sincere thanks to University Grants Commission for providing fellowship to the author PB during the course of this study.

Disclosure statement

There are no competing interest.

Funding

The author(s) reported there is no funding associated with the work featured in this article.

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